

A Comprehensive Analysis on Skin Cancer Classification Using Transfer Learning

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Abstract— Skin cancer, a prevalent and potentially life-threatening disease, necessitates accurate and timely diagnosis for effective treatment. In addressing this critical healthcare challenge, this research proposes a novel approach to skin cancer classification. The study identifies the need for enhanced diagnostic accuracy and introduces a solution by integrating the efficiency of transfer learning, specifically utilizing the EfficientNet V2 architecture. The proposed methodology leverages pre-trained convolutional neural networks (CNNs) to extract high-level features from dermatological images, which are subsequently input into a sophisticated classifier, forming a robust framework for precise skin cancer classification. The research employs a diverse dataset comprising 3297 images sourced from the publicly available Skin Cancer ISIC archive dataset, ensuring comprehensive coverage of benign and malignant skin lesions. During the training phase, the proposed EfficientNet V2 transfer learning approach outperforms conventional methods, achieving an impressive accuracy of 84%, surpassing both Inception V3 (82%) and a generic CNN model (81%). Interpretability analysis is conducted to elucidate the decision-making process of the EfficientNet V2 model, providing insights into the key features influencing its predictions. This transparency is crucial for building trust in the model's recommendations and facilitating its seamless integration into clinical workflows. The EfficientNet V2 transfer learning approach emerges as a promising and effective solution for advancing skin cancer classification systems, contributing significantly to the ongoing efforts to improve diagnostic accuracy in dermatology.

Keywords— Skin cancer, Convolution Neural Network (CNN), Inception, EfficientNet V2, Transfer Learning.

I. INTRODUCTION

Skin cancer, a prevalent and potentially life-threatening condition, arises when the skin cells undergo malignant transformations, often due to prolonged exposure to ultraviolet (UV) radiation from the sun or artificial sources. It encompasses various types, with the most common being basal cell carcinoma, squamous cell carcinoma, and melanoma. As one of the most diagnosed forms of cancer globally, understanding its prevalence and devising effective methods for early detection and classification are crucial in improving outcomes for individuals at risk.

According to the World Health Organization (WHO), skin cancers comprise the most frequently diagnosed groups of cancers globally, surpassing 1.5 million new cases in 2020 alone [1]. This alarming statistic underscores the pervasive nature of this health issue and its impact on individuals worldwide. In the same year, an estimated 325,000 new cases of melanoma, a particularly aggressive form of skin cancer, were diagnosed globally, resulting in 57,000 deaths [2]. These numbers illuminate the urgent need for effective strategies to address the rising incidence rates and mortality associated with skin cancer, emphasizing the critical role of advancements in early detection and classification methodologies.

In recent years, machine learning, and particularly transfer learning, has emerged as a promising tool in the field of medical image analysis, including the classification of skin cancer. Transfer learning leverages pre-trained neural networks on large datasets to improve the performance of models when applied to a specific task, such as skin cancer classification. By transferring knowledge learned from diverse datasets, these models can efficiently identify patterns and features indicative of different skin cancer types. This approach holds immense potential in enhancing the accuracy and speed of diagnosis, ultimately contributing to more effective and timely interventions for individuals at risk of or affected by skin cancer.

II. LITERATURE REVIEW

The intricate nature of skin cancer poses formidable challenges to both medical science and individuals affected by the disease. The profound impact of skin cancer on physical well-being and overall quality of life highlights the critical need for accurate and timely detection [3]. This literature review delves into the current understanding of skin cancer, its varied manifestations, and the evolving strategies for diagnosis and treatment.

This paper [4] proposes an innovative approach, including contrast stretching and the OTSU thresholding algorithm, to enhance skin lesion classification and segmentation, providing efficient and economical solutions for timely diagnosis. From paper [5], explores the pivotal role of early detection in addressing skin cancer risks, presenting the

effectiveness of Artificial Neural Network (ANN) and Convolutional Neural Network (CNN) through deep learning techniques. The study underscores the potential impact of these advanced technologies in enhancing accuracy and efficiency, emphasizing their significance in automating skin cancer detection for improved treatment outcomes. Paper [6] introduces a novel ensemble-based CNN architecture, leveraging both pre-trained and data-specific models, along with metadata, to enhance classification accuracy, demonstrating superior performance on a dataset of 33,126 dermoscopic images compared to benchmark methods. In paper [7], addressing Melanoma skin cancer, this research focuses on Segmentation and Classification, utilizing a Color Layout Filter model and achieving an optimal accuracy of 90.96%, precise 91%, and recall 0.91. The study recommends the efficient Attribute Selected Classifier, incorporating image enhancement techniques for improved outcomes in Melanoma detection. Highlighting the crucial role of computer vision in early disease detection, particularly in fatal conditions like skin cancer, this paper introduces two effective methods utilizing k-nearest neighbor and an optimized AlexNet model for dermoscopic image classification. The experiments, conducted on 4000 images from the ISIC archive dataset, demonstrate the superiority of the proposed methods over various machine learning and deep learning approaches in skin cancer detection [8]. Addressing the urgency of early skin cancer detection, this study proposes an automated classification method utilizing fine-tuned deep learning models, including VGG16, ResNet50, and ResNet101. Through innovative techniques such as data augmentation, generative adversarial networks (GANs), and conditional GANs, the study aims to enhance skin lesion classification, demonstrating improved performance compared to models developed with unbalanced datasets [9].

The literature review underscores the transformative potential of technological advancements, particularly in the context of skin cancer detection.

III. METHODOLOGY

A. Working Flowchart

The process of skin cancer classification involves a series of systematic steps to ensure accurate and reliable results. Initially, the collected dermoscopic images undergo various preprocessing methods, such as resizing, normalization, and noise reduction, to enhance the quality and consistency of the data. Following this, feature extraction is employed to identify relevant patterns and characteristics within the images. Extracted features serve as the discriminative elements for distinguishing between different types of skin lesions. The dataset is partitioned into training and testing sets, allowing the model to learn from a subset of the data and assess its performance on unseen examples. Transfer learning, a powerful technique in this context, involves utilizing pre-trained neural networks on large image datasets to leverage knowledge gained from unrelated tasks. Fine-tuning is then performed on the skin cancer dataset to adapt the model to the specific nuances of dermoscopic images. The training

process refines the model's parameters through iterative adjustments, optimizing its ability to accurately classify skin lesions.

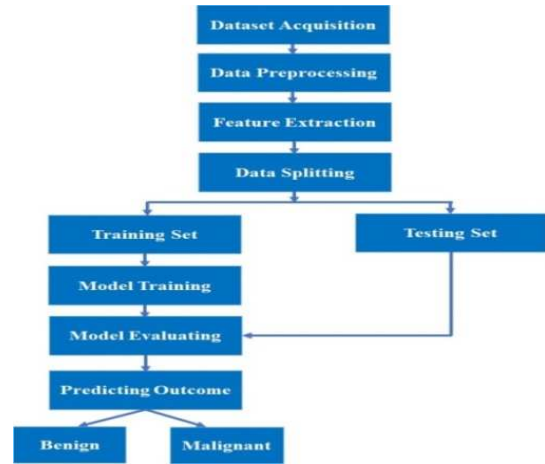


Fig. 1 Proposed flowchart

Once the model is trained, it can be applied to predict outcomes on new, unseen images. These predictions provide valuable insights into the likelihood of a given skin lesion being malignant or benign, aiding clinicians in making informed decisions about patient care. The integration of transfer learning and advanced machine learning techniques in skin cancer classification represents a significant stride towards more efficient and precise diagnostic tools in the realm of dermatology.

B. Data Preprocessing

In the preprocessing phase of the skin cancer dataset, meticulous steps are undertaken to ensure the data is suitably formatted for subsequent analysis. One crucial step involves image resizing, where images are uniformly adjusted to a standardized dimension [10]. This not only facilitates computational efficiency but also ensures that the model is invariant to variations in image size, promoting robust generalization. Following resizing, normalization is applied to standardize pixel values across the dataset, mitigating the influence of differing illumination conditions and enhancing the model's ability to discern meaningful patterns. Normalization transforms pixel values to a common scale, typically ranging between 0 and 1, thereby promoting convergence during training and fostering a more stable and effective learning process. These preprocessing measures collectively contribute to a well-conditioned skin cancer dataset, setting the stage for the subsequent development of accurate and reliable machine learning models for skin cancer classification.

C. Data Augmentation

The augmentation of the dataset was integral to enhancing its size and augmenting the generalization capabilities of the model. To introduce diversity into the training dataset, a suite of data augmentation techniques was judiciously applied,

including random rotations, flips, and zooms. These transformations served to introduce subtle variations, enabling the model to learn more robust features and generalize effectively to unseen instances. Furthermore, a critical step involved the conversion of images into a standardized color space, such as RGB.

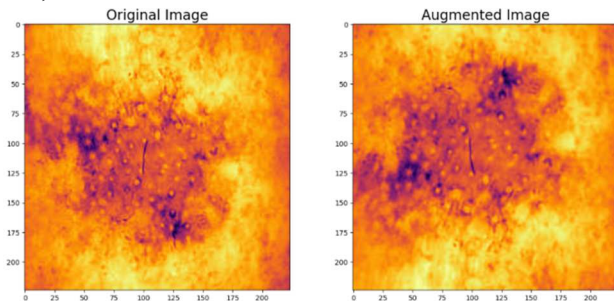


Fig. 2 Augmented skin cancer image

This transformation ensures consistency in feature extraction across the dataset, as it harmonizes the representation of images, regardless of their original color space [11]. This process involves translating images from one color space to another, with the objective of maintaining the visual fidelity of the original while aligning it with a standardized color representation. By meticulously employing these techniques, the dataset was not only expanded but also refined to bolster the model's capacity for accurate and adaptable skin cancer classification.

D. Transfer Learning

Transfer learning is a machine learning technique that utilizes knowledge gained from solving one problem and applies it to a different but related task. In the context of skin cancer detection, our approach involved employing pre-trained models such as efficientNet, CNN (Convolutional Neural Network), and Inception. These models had initially been trained on large datasets for general image recognition tasks [12]. By transferring the learned features and weights from these models to the specific task of skin cancer detection, we were able to significantly enhance the efficiency and accuracy of our system. This approach is particularly beneficial when the dataset for the target task is limited, as it taps into the wealth of information learned from broader datasets in related domains. The combination of efficientNet, CNN, and Inception showcased the adaptability and versatility of transfer learning in addressing complex medical image analysis tasks like skin cancer detection.

1) EfficientNet V2

The model configuration presented is a neural network architecture designed for skin cancer detection, utilizing the EfficientNetV2_B0 as its base. The architecture consists of several layers, including the EfficientNetV2_B0 layer, a dropout layer to prevent overfitting, and two dense layers for classification. The EfficientNetV2_B0 layer serves as the backbone, extracting relevant features from the input data. The model has a total of 6,083,538 parameters, with 164,226 being trainable and 5,919,312 non-trainable. The trainable parameters are those that can be adjusted during the training

process to optimize the model for the specific task at hand. This configuration reflects a balance between model complexity and efficiency, with a focus on leveraging pre-trained weights from EfficientNetV2_B0 to enhance the performance of skin cancer detection. The small number of trainable parameters indicates a fine-tuning approach, where the model adapts to the nuances of the target task while benefiting from the knowledge captured by the pre-trained base model.

2) Convolutional Neural Network (CNN)

Convolutional neural network (CNN) architecture designed for a skin cancer detection task. The network consists of multiple layers, including Conv2D convolutional layers followed by max-pooling layers for spatial down-sampling. The convolutional layers are designed to capture hierarchical features from the input images, while max-pooling reduces spatial dimensions, aiding in feature extraction [13]. The architecture employs two sets of convolutional layers and max-pooling operations, followed by a flatten layer to transform the 3D output into a 1D array. Subsequently, a dropout layer is incorporated to mitigate overfitting. The model concludes with two dense layers for classification, with the final layer producing a binary output for skin cancer detection. The total number of parameters in this model is 2,881,314, all of which are trainable. This configuration indicates a moderate-sized model with a focus on capturing relevant image features for effective skin cancer classification. The absence of non-trainable parameters implies that the model is not utilizing pre-trained weights, allowing it to adapt specifically to the nuances of the skin cancer detection task during training.

3) Inception V3

The presented model configuration showcases a smart utilization of the InceptionV3 architecture for skin cancer detection. InceptionV3, a powerful pre-trained model, serves as the backbone for feature extraction, bringing in a vast wealth of knowledge acquired from general image recognition tasks. The subsequent layers, including a dropout layer for regularization and two dense layers for classification, contribute to fine-tuning the model to the specific requirements of skin cancer detection. The intelligent aspect of this configuration lies in the judicious choice of leveraging the pre-trained InceptionV3 model, which dramatically reduces the number of trainable parameters to 128,514. This not only saves computational resources but also allows the model to adapt efficiently to the nuances of the target task. The non-trainable parameters, totaling 23,853,833, underscore the substantial knowledge transfer from the InceptionV3 model. This approach reflects a smart and resource-efficient strategy in harnessing the power of pre-trained models for effective and accurate skin cancer detection.

IV. RESULT ANALYSIS

A. Dataset Description

The skin cancer dataset utilized in our study was sourced from the ISIC archive platform [14], representing a publicly accessible repository of images specifically curated for skin

cancer detection. Comprising a total of 3,297 images, this dataset serves as a valuable resource for training and evaluating machine learning models. To ensure robust assessment, a testing sample of 660 images and a training sample of 2,637 images were carefully partitioned. The dataset encompasses a diverse range of skin lesion images, capturing various manifestations of skin cancer, which is crucial for the model to learn and generalize effectively.

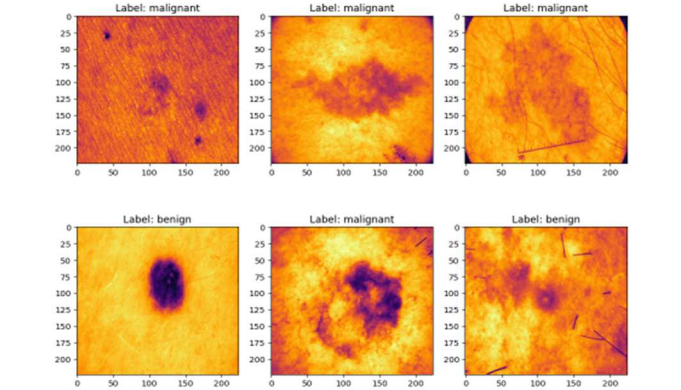


Fig. 3 Skin cancer dataset sample images

The availability of a substantial number of images facilitates the development of models capable of recognizing subtle patterns and features indicative of skin cancer, contributing to the advancement of diagnostic capabilities in the realm of dermatology. The utilization of such a comprehensive and openly accessible dataset enhances the transparency, reproducibility, and collaborative potential of research endeavors in skin cancer detection.

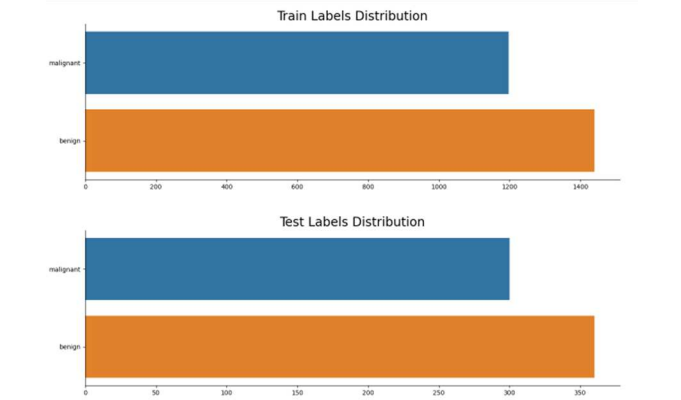


Fig. 4 Dataset distribution

The skin cancer dataset exhibits a balanced data distribution, ensuring that the training and testing sets encompass a proportional representation of both malignant and benign cases (Shows in fig. 4). This balanced distribution aids in preventing model bias and promotes robust performance across various skin cancer types.

B. Performance Analysis

We employed three distinct transfer learning models—CNN, EfficientNet V2, and Inception V3—to assess their individual performances. The results are succinctly presented

in Table I, showcasing the effectiveness of each model in the context of our study.

TABLE I. PERFORMANCE COMAPRISON

Algorithms	Accuracy	Precision	Recall	F1Score	Matthew's Corr. Coeff.
CNN	0.81	0.83	0.81	0.81	0.64
EfficientNet V2	0.84	0.85	0.84	0.84	0.69
Inception V3	0.82	0.82	0.82	0.82	0.64

In Fig. 5, the loss and accuracy curves for the CNN, EfficientNet V2, and Inception V3 models are presented, with EPOCHS set to 10 and a BATCH_SIZE of 32. The SEED value is established as 42, providing reproducibility in results. Additionally, TF_SEED is set to 768, ensuring consistency in TensorFlow operations. The HEIGHT and WIDTH of the images used for training and evaluation are both standardized to 224. These parameters play a crucial role in shaping the performance metrics displayed in the curves, offering a comprehensive view of the models' behavior over the specified number of epochs and batch size.

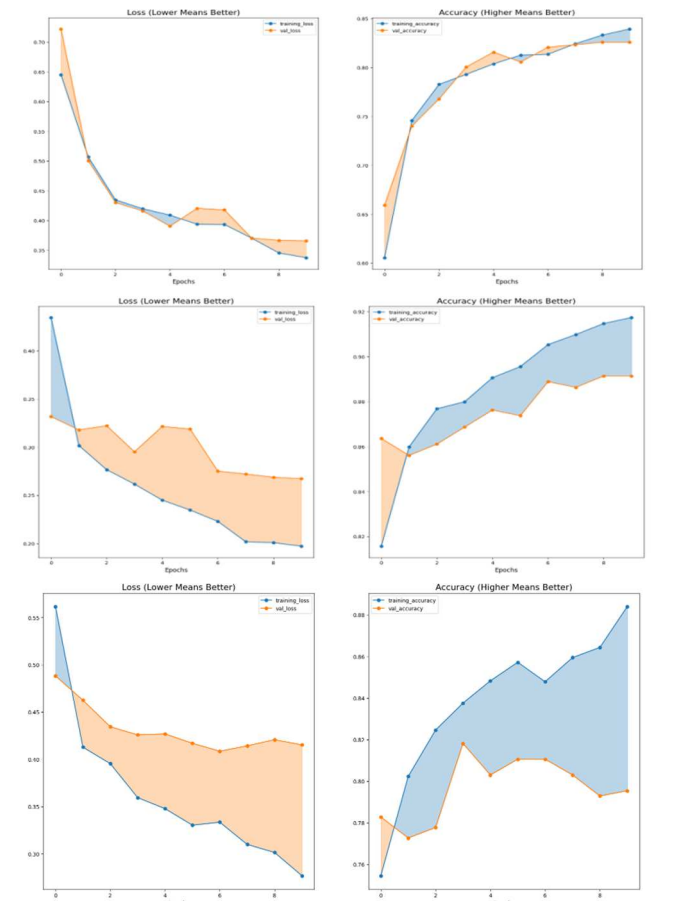


Fig. 5 Loss and accuracy curve of (a) CNN, (b) EfficientNet V2, and (c) Inception V3

In Fig. 6, a bar chart is presented, offering a visual comparison of the performance of different models. This graphical representation succinctly illustrates the key metrics or evaluation criteria, such as accuracy, precision, recall, or F1 score, across the CNN, EfficientNet V2, and Inception V3 models.

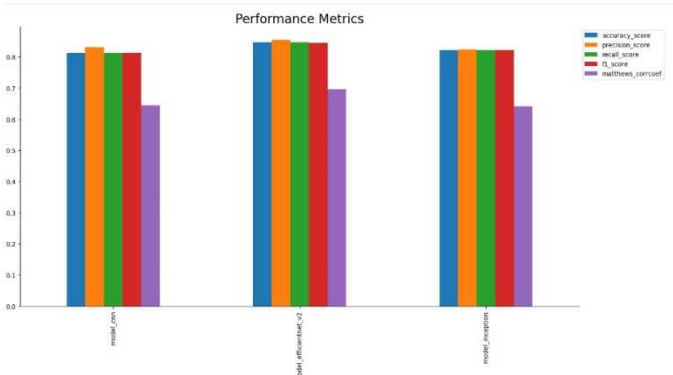


Fig. 6 Performance visualization

The confusion matrices serve as insightful visualizations in evaluating the classification performance of the models. Displayed in Fig. 7, 8, and 9, these matrices provide a detailed breakdown of true positive (TP), true negative (TN), false positive (FP), and false negative (FN) predictions for the CNN, EfficientNet V2, and Inception V3 models, respectively.

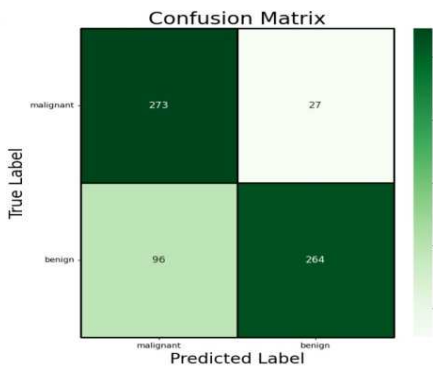


Fig. 7 CNN algorithm confusion matrix

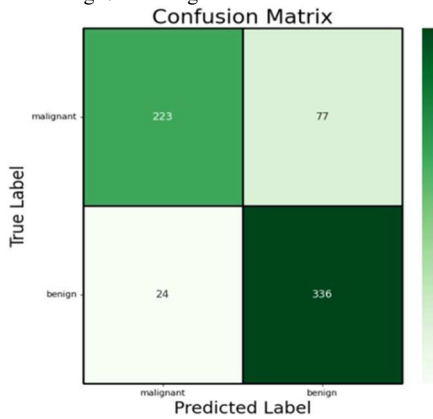


Fig. 8 EfficientNet V2 algorithm confusion matrix

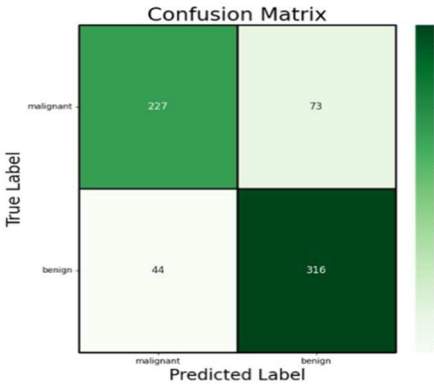


Fig. 9 Inception V3 algorithm confusion matrix

Deep learning models, such as convolutional neural networks (CNNs) and transfer learning architectures like EfficientNet and InceptionV3, have demonstrated remarkable success in recognizing patterns and features indicative of skin cancer.

C. Predicting Outcomes

The CNN, EfficientNet V2, and Inception V3 models predict outcomes by analyzing dermoscopic images and identifying patterns indicative of skin cancer, providing crucial insights for early detection and intervention. Harnessing the capabilities of these advanced machine learning algorithms, the models offer rapid and accurate predictions, aiding healthcare professionals in making informed decisions and improving patient outcomes.

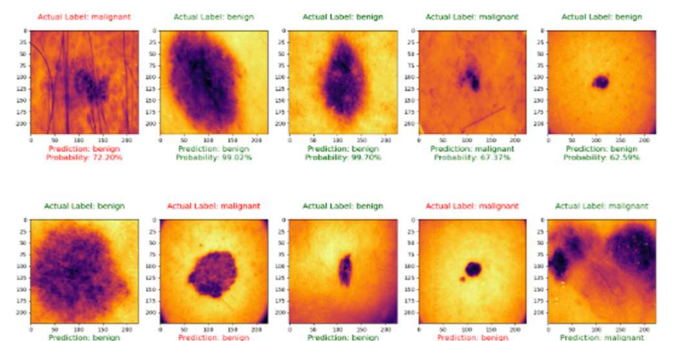


Fig. 10 CNN algorithm predictions

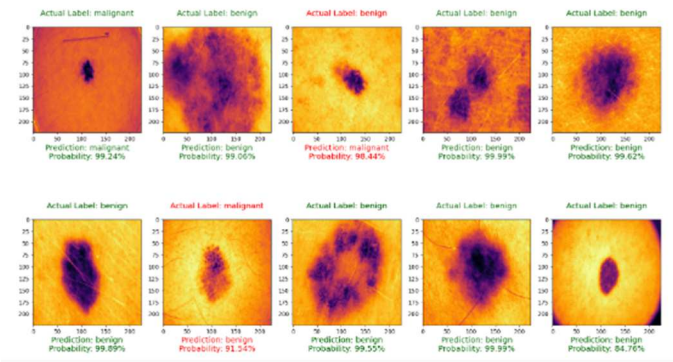


Fig. 11 EfficientNet V2 algorithm predictions

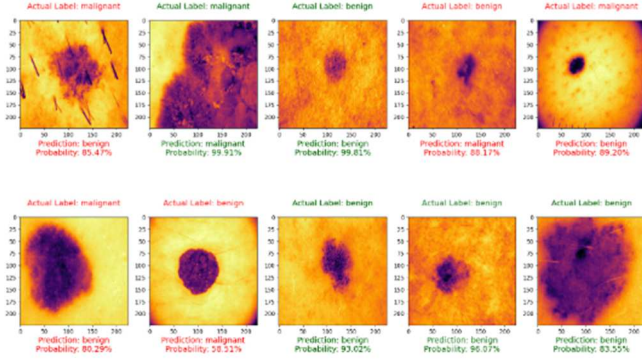


Fig. 12 Inception V3 algorithm predictions

The CNN, EfficientNet V2, and Inception V3 models excel in predicting outcomes from dermatoscopic images. Notably, the EfficientNet V2 model demonstrates superior performance, showcasing heightened accuracy and efficiency in identifying patterns indicative of skin cancer, as illustrated in Fig. 10, 11, and 12. Table II shows the comparative analysis between the proposed system and the existing system.

TABLE II. COMPARATIVE ANALYSIS

Authors	Algorithms	Performance
Gajera et al. [15]	DenseNet-121 with MLP	80.47% (ISIC 2016), 81.16% (ISIC 2017),
Pradhumn et al. [16]	MobileNet	80.81% (HAM10000 ISIC dataset)
Our Proposed	EfficientNet V2	84% (ISIC dataset)

V. CONCLUSION AND FUTURE WORK

Our research underscores the significant advantages of employing the EfficientNet V2 transfer learning approach during the training phase, surpassing conventional methods. With an impressive accuracy of 84%, the EfficientNet V2 model outperforms both Inception (82%) and a generic CNN model (81%). Moreover, interpretability analysis sheds light on the decision-making process of the EfficientNet V2 model, offering insights into the critical features that influence its predictions. This transparency proves pivotal for establishing trust in the model's recommendations, fostering its seamless integration into clinical workflows.

For future work, further refinement and fine-tuning of the models can be explored to optimize their performance on diverse skin lesion datasets. Continuous updates to models with evolving datasets and the integration of emerging technologies, such as multispectral imaging or 3D analysis, present exciting avenues for advancing the field of automated skin cancer detection, ultimately contributing to improved patient care and outcomes.

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